

Royal College of Arts, Science and Commerce (Autonomous)

Affiliated to University of Mumbai

Program: Bachelor of Science

Course: Computer Science

Syllabus for Semester: V and VI

Syllabus for Undergraduate Programme as per

National Education Policy (NEP-2020) with effect from the academic year 2026-2027

NEP Credit Structure for Computer Science Department

Level	Sem	Major		Minor	OE	VSC	SEC	AEC	IKS	VEC	OJT/FP /RP/CC /CEP	Cumulative Credits	
		DSC	DSE										
4.5	I	4TH+2PR 6		2TH	2TH+ 2TH	2PR	2PR	2TH	2TH	2TH		22	**UG Certificate Cumulative Credit:44**
	II	4TH+2PR 6		2TH	2TH+ 2TH	2PR	2PR	2TH		2TH	2 CC	22	
Exit Option: Award of UG Certificate in Major with 40-44 Credits and an Additional 4 Credits Core NSQF Course / Internship OR Continue with Major and Minor													
5	III	8TH+2PR 10		2TH+2PR 4	2TH		2PR	2TH			2 CC	22	**UG Diploma Cumulative Credit:88**
	IV	8TH+2PR 10		2TH+2PR 4	2TH		2PR	2TH			2 CEP	22	
Exit Option: Award of' UG Diploma in Major and Minor with 80-88 Credits and an Additional 4 Credits Core NSQF Course /Internship OR Continue with Major and Minor													
5.5	V	8TH+2PR 10	2	2TH+2TH 4		2					4 FP	22	**UG Degree Cumulative Credit:132**
	VI	8TH+2PR 10	2	2TH+2TH 4		2					4 OJT	22	
Total		**52**	**4**	**20**	**12**	**8**	**8**	**8**	**2**	**4**	**14**	**132**	

Programme Outcomes (POs) for Computer Science

	On completing B.Sc. Computer Science, the student will be able to:
PO1	Develop understanding and knowledge of the fundamental theories, applications, technologies in computer science.
PO2	Understanding and applying the knowledge of networking, web design, security, cloud computing, IoT.
PO3	Understanding of best practices, standards to develop user interactive and abstract applications.
PO4	Demonstrate proficiency in using current techniques, skills, and tools necessary for computing practice.
PO5	Apply concepts, principles and theories relating to computer science to new situations.
PO6	Apply standard software engineering practices and strategies in real-time software project development.
PO7	Apply technical skills to formulate & design solutions to solve real world problems.
PO8	Pursue higher studies of specialization and to take up technical employment.
PO9	Develop Work Ethics , Communication Skills to present their work effectively and coherently.
PO10	Engage in independent and life-long learning in the background of the rapidly changing IT industry.

Programme Specific Outcomes (PSOs) for Computer Science

	On completing B.Sc. Computer Science, the student will be able to:
PSO1	Demonstrate proficiency in different computing paradigms needed for a proper understanding of computer science.
PSO2	Develop and implement software solutions to effectively address real-world problems in business, science, and social contexts.
PSO3	Build up programming, analytical and logical thinking abilities.
PSO4	Design, develop and analyze the use of the Information security tools and technique for providing security.
PSO5	Demonstrate programs in various languages to enhance computer and internet capabilities.
PSO6	Apply the standard software engineering practices and strategies in real time software project development.

Semester –V				
Course Code	**Course Type**	**Course Title**	**Credits**	**Lectures /Week**
RUCSMJ501	Major – I	Artificial Intelligence & Machine Learning	4	4
RUCSMJ502	Major – II	Software Project Management	2	2
RUCSMJ503	Major – III	Indian Knowledge Computational Thinking	2	2
RUCSMJP5	Major Practical	Computer Science Practical 5	2	4
RUCSDSE5041	Elective – I*	Information Retrieval	2	2
RUCSDSE5042	Elective – II*	Linux Server Administration	2	2
RUCSMN505	Minor	Cloud Computing & Web Services	2	2
RUCSMNP505	Minor	Cloud Computing & Web Services- Practical	2	4
RUCSVSC506	VSC	Ethical Hacking & Cyber Forensics	2	4
RUCSFP507	FP	Field Project	4	-

*** DSE - Discipline Specific Elective subject will be offered any one from Elective – I and Elective – II**

Semester –VI				
Course Code	**Course Type**	**Course Title**	**Credits**	**Lectures /Week**
RUCSMJ601	Major – I	Cryptography & Network Security	4	4
RUCSMJ602	Major – II	Software Testing & Quality Assurance	4	4
RUCSMJP6	Major Practical	Computer Science Practical 6	2	4
RUCSDSE6031	Elective – I	Game Programming	2	2
RUCSDSE6032	Elective – II	Wireless Sensor Networks	2	2
RUCSMN604	Minor	Data Mining & Warehousing	2	2
RUCSMNP604	Minor	Data Mining & Warehousing - Practical	2	4
RUCSVSC605	VSC	NO-SQL	2	4
RUCSOJT606	OJT	On Job Training	4	-

*** DSE - Discipline Specific Elective subject will be offered any one from Elective – I and Elective – II**

Course/ Paper Title	**Artificial Intelligence & Machine Learning**
Course offered as	**Major I**
Course Code	**RUCSMJ501**
Semester	**V**
No. of Credits	4 (60 hours)
No. of lecture Hours/week	4

Sr No.	Course Learning Objectives:
CLO1	Understand fundamental concepts of Artificial Intelligence, intelligent agents, and search strategies.
CLO2	Apply knowledge representation, logical reasoning, and planning techniques in AI problems.
CLO3	Analyze and implement supervised machine learning models for classification and regression tasks.
CLO4	Evaluate advanced AI techniques including ensemble learning, clustering, probabilistic models, and reinforcement learning.
CLO5	Design and develop AI based solutions using appropriate tools and frameworks for real-world applications.

Course Outcome

	On completing the course, the student will be able to:
CO1	Explain AI concepts and apply search algorithms to solve problem-solving tasks.
CO2	Apply logical reasoning, knowledge representation, and planning techniques to model intelligent systems.
CO3	Implement and analyze supervised learning algorithms for classification and regression problems
CO4	Evaluate advanced learning methods including ensemble models, clustering, probabilistic models, and reinforcement learning.
CO5	Develop and implement AI models to solve real-world problems using suitable algorithms, tools and evaluation techniques.

Detailed Syllabus

Module	Title with content	No. of lectures
I	**What Is AI:** History, Components of AI Program, Foundations of AI, Sub-areas of AI, Applications of AI, Current trends in AI. **Intelligent Agents:** Agents and Environments, the concept of rationality, the nature of environment, the structure of Agents, Types of Agents, Learning Agent. **Problem Solving by searching:** Problem-Solving Agents, Uninformed Search Methods: Breadth First Search (BFS), Depth First Search (DFS), Depth Limited Search, IDDFS, Informed Search Methods: Greedy best first Search, A* Search, Memory bounded heuristic Search.	**15**
II	**Knowledge Representation and Reasoning:** Knowledge based Agents, Brief Overview of propositional logic, First Order Logic: Syntax and Semantic, Inference in FOL, Forward chaining, backward Chaining.	**15**

	Knowledge Engineering in First-Order Logic, Unification, Resolution, Uncertain Knowledge and Reasoning: Uncertainty, Representing knowledge in an uncertain domain, The semantics of Bayesian (Belief) Networks **Planning and Learning**: The planning problem, Planning with state space search, Partial order planning, Hierarchical planning, Conditional Planning.	
III	**Machine Learning**: Introduction, Types of Machine Learning, Issues in Machine Learning, Application of Machine Learning, Steps in developing a Machine Learning Application, Parametric vs Non-Parametric Models, Classification vs Regression, Overfitting vs Underfitting **Supervised Learning:** KNN, SVM, Naive Bayes **Learning with Regression:** Linear Regression, Multivariate Linear Regression, Logistic Regression. **Learning with trees:** Constructing Decision Trees, Gini Index, Entropy, Information Gain **Understanding Artificial neural network**	**15**
IV	**Ensemble Learning** Understanding Ensembles, K-fold cross validation, Boosting, Stumping, Bagging, Random Forest **Unsupervised Learning**: Concept of Unsupervised learning, Clustering & its types Association Rule Mining, Dimensionality Reduction: Principal Component Analysis, Linear Discriminant Analysis **Probabilistic models:** Statistical Learning, Learning with Hidden Variables: The EM Algorithm, Hidden Markov Model **Reinforcement learning:** Concept of Reinforcement learning, Q-Learning	**15**

References:

1. Artificial Intelligence: A Modern Approach, Stuart J. Russell and Peter Norvig, 4th Edition, Pearson Education, 2020.
2. A First Course in Artificial Intelligence, Deepak Khemani, McGraw-Hill Education (India), 2013.
3. Artificial Intelligence, Saroj Kaushik, Cengage Learning India, 2011.

Additional References:

1. Machine Learning, Tom M. Mitchell, McGraw-Hill Education, 1997.
2. Introduction to Machine Learning, Ethem Alpaydın, 3rd Edition, MIT Press, 2014.
3. Machine Learning: An Algorithmic Perspective, Stephen Marsland, 2nd Edition, CRC Press, 2015.

Course/ Paper Title	**Software Project Management**
Course offered as	**Major II**
Course Code	**RUCSMJ502**
Semester	**V**
No. of Credits	**2**
No. of lecture Hours/week	**2**

Sr No.	**Course Learning Objectives:**
CLO1	Understand the Concepts of Software Engineering.
CLO2	Apply software process models to project scenarios
CLO3	Analyze software requirements and create UML-based system models
CLO4	Evaluate and implement appropriate project management practices to ensure successful software project delivery.
CLO5	Apply knowledge of human resource management techniques, including team development, conflict resolution, and communication management.

Course Outcome

	On completing the course, the student will be able to:
CO1	Explain the fundamental concepts of software engineering, software process models, and professional software development practices
CO2	Apply suitable software process models (Waterfall, Incremental, RAD, Agile, Spiral, etc.) to different project scenarios.
CO3	Analyze software requirements and develop system models using UML diagrams.
CO4	Demonstrate knowledge of project management fundamentals, including project initiation, scope management, scheduling, cost estimation, and monitoring.
CO5	Apply project time and cost management techniques and cost control methods.

Detailed Syllabus

Module	**Title with content**	**No. of lectures**
I	**Introduction:** The Nature of Software, Software Engineering, Professional Software Development, Waterfall Model, Incremental, RAD Models Evolutionary Process Models: Prototyping, Spiral and Concurrent Development Model Specialized Models: Component based, Aspect Oriented development, The Unified Process Phases, Agile Development-Agility, Agile Process, Extreme Programming. **Requirement Analysis:** Requirements Engineering, Eliciting Requirements, SRS Validation, Components of SRS, Characteristics of SRS, **System Modelling:** Object-oriented design using the UML - Class diagram, Object diagram, Use case diagram, Sequence diagram, Collaboration	**15**

	diagram, State chart diagram, Activity diagram, Component diagram, Deployment diagram. **System Design:** System/Software Design, Architectural Design, Low Level Design Coupling and Cohesion, Design Specifications, Verification for Design, Monitoring and Control for Design.	
II	**Introduction to Project Management**: Definition and characteristics of a project, Importance of project management, Project management processes and knowledge areas **Project Selection, Initiation and scope Management:** Project selection criteria and methods, Project initiation and charter development, Stakeholder identification and analysis, Scope planning and definition, Work Breakdown Structure (WBS) development, Scope verification and change control. **Project Time & Cost Management:** Activity definition and sequencing, Estimating activity durations and resources, Developing the project schedule, Schedule control and monitoring, Cost estimation techniques, Budget development and monitoring, Earned Value Management (EVM), Cost control and analysis. **Project Resource & Procurement Management:** Team development and roles, Staffing, training, and motivation, Conflict resolution and communication management, Procurement planning and contract types, Solicitation, source selection, and contract administration, Vendor management and relationship building. Project integration processes and activities, Change management and project closure, Lessons learned and knowledge transfer	**15**

References:

1. Software Engineering, A Practitioner's Approach, Roger S, Pressman, 2019
2. Software Engineering: principles and Practices, Deepak Jain, OXFORD University Press, 2008
3. Project Management for Business and Technology, 3rd edition, Pearson Education. John M. Nicholas, 2000
4. Information Technology Project Management, by Jack T. Marchewka, 4th Wiley India 2013.
5. A Guide to the Project Management Body of Knowledge, Sixth Edition 6th Edition, Project Management Institute

Additional References:

1. Software Engineering, Ian Sommerville, Pearson Education, 2017
2. Fundamentals of Software Engineering, Fourth Edition, Rajib Mall, PHI, 2018
3. Software Engineering, first Edition, Pramitha Santhumayor, Techneo Publication,2025
4. Introduction to Software Project Management by Adolfo Villafiorita · 2016, CRC press, e book format.

Course/ Paper Title	**Indian Knowledge Computational Thinking**
Course offered as	**Major**
Course Code	**RUCSMJ503**
Semester	**V**
No. of Credits	**2 (30 Hrs)**
No. of lecture Hours/week	**2**

Sr No.	Course Learning Objectives:
CLO1	To introduce the foundational principles of Indian Knowledge Systems and their connection to computational thinking.
CLO2	Understand the principles of logic, reasoning, and algorithmic thinking from classical Indian texts.
CLO3	To demonstrate Sanskrit as a formal knowledge representation language using the rule-based grammar of Panini.
CLO4	Understand the application of Paninian grammar principles in Natural Language Processing.
CLO5	To examine Ayurvedic rule-based diagnostic principles for modeling expert decision-support systems.
CLO6	To develop the ability to design and implement knowledge representation models derived from traditional Indian texts for applications in AI and modern computing.

Course Outcome

	On completing the course, the student will be able to:
CO1	Explain the foundational principles of Indian Knowledge Systems (IKS) and their role in computational thinking and algorithmic reasoning.
CO2	Analyze logic and reasoning from Nyaya Sutras and algorithmic principles in Sulba Sutras for application in modern computing.
CO3	Describe Sanskrit as a knowledge representation language and apply Panini's rule-based grammar and sutra system to understand formal grammar, morphological analysis, and parsing.
CO4	Implement basic NLP tools using Paninian grammar and modern NLP libraries for tasks such as morphological analysis, parsing, and machine translation.
CO5	Apply Ayurvedic dosha-based rules to develop diagnostic decision-support systems and understand rule-based expert systems.
CO6	Design knowledge representation structures (ontologies, knowledge graphs) from traditional texts and evaluate their application in AI and modern computing systems.

Detailed Syllabus

Module	Title with content	No. of lectures
I	**Foundations of IKS and Computational Thinking:** Overview of Indian Knowledge Systems and scientific traditions, Computational thinking in ancient India ,***Acharya Pingalas -Mathematical logic and Binary System and pattern***. logic and reasoning in Nyaya Sutras, Algorithmic thinking in Sulba Sutras , Knowledge organization in Vedas, Data Knowledge classification in Ayurveda.***Arthashastra-Cryptography*** **Sanskrit, Formal Grammar, and NLP:** ***Marathi and Hindi Language using for NLP***.Sanskrit as a knowledge representation language, Rule-based grammar of Panini and formal language theory, Sutra system as production rules (IF–THEN rules).	**15**
II	**Case Studies & Applications of IKS in Modern Computing:** Paninian Linguistics for NLP: Sandhi rules for phonological processing and token normalization, Samasa for morphological decomposition, Karaka theory for syntactic role labelling and dependency parsing, Rule-based morphological analyzer using Paninian principles for NLP, Lemmatization and word formation using sutra-based rules, Syntax parsing using karaka relations and dependency structures. **Design an NLP** tool using Paninian grammar with NLP libraries, case studies on Ayurvedic Diagnostic Systems by applying Ayurvedic dosha-based rules. Machine Translation for Sanskrit and Indian Languages.	**15**

References:

1. Introduction to Indian Knowledge System: Concepts and Applications by B. Mahadevan, M. A. Balasubramaniam, and K. Gayathri,2022 , PHI Publisher.
2. A Guide To Panini An English Summary Of PaniniS Aphorisms On Sanskrit Grammar by S. Vencatavaradiengar, Gyan Publishing House

Additional References:

1. Speech and Language Processing by Dan Jurafsky and James H. Martin, Pearson Publisher
2. Foundations of Statistical Natural Language Processing — Christopher D. Manning, Hinrich Schutze
3. Bansal, G. (2025). Indian Knowledge System: Integrating Heritage with Engineering. Deep Science Publishing. https://doi.org/10.70593/978-93-49307-29-2

Course/ Paper Title	**Computer Science Practical 5**
Course offered as	**Major Practical**
Course Code	**RUCSMJP5**
Semester	**V**
No. of Credits	2 (60hrs)
No. of lecture Hours/week	2

Sr. No.	**Course Learning Objectives of Module I & II**
CLO1	Understand and explain fundamental concepts of search strategies, machine learning, and data mining techniques.
CLO2	Apply uninformed and informed search algorithms to solve real-world problems.
CLO3	Implement supervised learning algorithms for classification and regression tasks.
CLO4	Analyze model performance using appropriate evaluation metrics and interpret the results effectively.
CLO5	Design and implement advanced techniques such as Clustering, Association Rule Mining, and Ensemble Learning methods.
CLO6	To understand and apply software engineering principles in the design and development of a real-world game-based or interactive system.
CLO7	To enhance analytical, modeling, and system design skills for building scalable and maintainable software systems.
CLO8	To understand behavioral, structural, and implementation views of a system using UML modeling techniques.
CLO9	To develop documentation skills required for professional software development life cycle (SDLC) processes.
CLO10	To model system requirements using UML diagrams.

Course Outcome

	On completing the course, the student will be able to:
CO1	Implement and compare uninformed and informed search algorithms for problem-solving applications.
CO2	Develop supervised learning models and evaluate their predictive performance.
CO3	Perform Association Rule Mining and compute support and confidence for generated rules.
CO4	Apply clustering techniques and dimensionality reduction methods for data analysis and visualization.
CO5	Implement and compare uninformed and informed search algorithms for problem-solving applications.
CO6	Implement and compare uninformed and informed search algorithms for problem-solving applications.
CO7	Identify, analyze, and define real-world problems and formulate clear problem statements for software system development.
CO8	Design system models using UML diagrams including Use Case, Class, Object, Sequence, Collaboration, State Chart, Activity, Component, and Deployment diagrams.
CO9	Demonstrate the ability to design structural, behavioral, and implementation views of a system following Software Development Life Cycle (SDLC) principles.
CO10	Enhance documentation, analytical, and system design skills required for professional software development projects.
CO11	Apply project planning and scheduling techniques by preparing Gantt charts and managing development timelines.
CO12	Identify, analyze, and define real-world problems and formulate clear problem statements for software system development.

Sr No.	**Module I: Artificial Intelligence & Machine Learning**
1	A) Implement the Breadth First Search algorithm to solve a given problem. B) Implement the Iterative Depth First Search algorithm to solve the same problem.
2	A) Implement the A* Search algorithm for solving a path finding problem. B) Implement the Recursive Best-First Search algorithm for the same problem.
3	Implement the Decision Tree Learning algorithm to build a decision tree for a given dataset. Evaluate the accuracy & Visualize the generated decision tree.

4	Implement the SVM algorithm for binary classification. Evaluate the performance of the SVM model on test data and analyse the results.
5	Implement the K-NN algorithm for classification or regression. Evaluate the accuracy or error of the predictions and analyze the results.
6	Implement the Naive Bayes' algorithm for classification. Evaluate the accuracy of the model on test data and analyze the results.
7	Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent item sets. Generate association rules from the frequent item sets and calculate their support and confidence.
8	To implement concept of the Clustering
9	To implement PCA/LDA
10	To implement Ensemble learning (bagging/boosting). Implement the Adaboost algorithm to create an ensemble of weak classifiers.

Sr No.	**Module II: Software Project Management**
1	Write down the problem statement for a suggested system of relevance. Prepare synopsis for the same.
2	Perform requirement analysis and develop Software Requirement Specification Sheet (SRS) for suggested system.
3	Prepare time Gantt Chart for suggested system.
4	Draw E-R diagram for suggested system.
5	Draw the user‘s view analysis for the suggested system: Use case diagram.
6	Draw the function oriented diagram: Data Flow Diagram (DFD) and Structured chart.
7	Draw the behavioral view diagram : State-chart diagram, Activity diagram
8	Draw the structural view diagram for the system: Class diagram, object diagram.
9	Draw the behavioral view diagram for the suggested system: Sequence diagram, Collaboration diagram
10	Draw the implementation and environmental view diagram: Component diagram, Deployment diagram.
	Note: Perform the above exercises for any one project given in the list of sample projects.**List of Sample Projects**: a) Student Result Management System b) Library Management System c) Inventory Control System d) Accounting System e) Fast Food Billing System f) Bank Loan System g) Blood Bank System h) Railway Reservation System i) Automatic Teller Machine j) Video Library Management System k) Hotel Management System l) Court case management system m) Share Online Trading

Course/ Paper Title	**Information Retrieval**
Course offered as	**Discipline Specific Elective - I**
Course Code	**RUCSDSE5041**
Semester	**V**
No. of Credits	**2 (30 hours)**
No. of lecture Hours/week	**2**

Sr No.	**Course Learning Objectives:**
CLO1	To understand the fundamentals, components, and applications of information retrieval systems.
CLO2	To apply indexing, storage, compression and term weighting techniques for efficient information storage and retrieval
CLO3	Implement and compare retrieval models such as Boolean, Vector Space, and Probabilistic models.
CLO4	To analyse advanced IR techniques including web crawling, link analysis, duplicate detection, and multilingual retrieval.
CLO5	To evaluate Information Retrieval systems using standard performance metrics and experimental methodologies.

Course Outcome

	On completing the course, the student will be able to:
CO1	Explain the concepts, components, and challenges of information retrieval systems.
CO2	Construct and manage inverted indexes with appropriate storage and compression techniques.
CO3	Apply and compare retrieval models such as Boolean, Vector Space, and Probabilistic models.
CO4	Analyze web IR techniques such as crawling, link analysis and multilingual retrieval.
CO5	Evaluate system performance using standard metrics.

Detailed Syllabus

Module	**Title with content**	**No. of lectures**

I	**Introduction to Information Retrieval (IR) systems:** Definition and goals of information retrieval, Components of an IR system, Challenges and applications of IR **Document Indexing, Storage, and Compression:** Inverted index construction and compression techniques, Document representation and term weighting, Storage and retrieval of indexed documents, **Retrieval Models: Boolean model:** Boolean operators, query processing, Vector space model: TF-IDF, cosine similarity, query-document matching, Probabilistic model: Bayesian retrieval, relevance feedback **Spelling Correction in IR Systems:** Challenges of spelling errors in queries and documents, Edit distance and string similarity measures, Techniques for spelling correction in IR systems	**15**
II	**Advanced Topics in Information Retrieval** **Web Information Retrieval:** Web search architecture and challenges, Crawling and indexing web pages. **Crawling and Near-Duplicate Page Detection**: Web page crawling techniques: breadth-first, depth-first, focused crawling, Near-duplicate page detection algorithms. **Link Analysis and its Role in IR Systems:** Web graph representation and link analysis algorithms, HITS, Page Rank. **Cross-Lingual and Multilingual Retrieval:** Challenges and techniques for cross-lingual retrieval, Machine translation for IR, Multilingual document representations and query translation **Performance Evaluation:** Evaluation metrics: precision, recall, F-measure, average precision, Test collections and relevance judgments, Experimental design and significance testing	**15**

References:

1. Ricardo Baeza-Yates and Berthier Ribeiro-Neto, ―Modern Information Retrieval: The Concepts and Technology behind Search, Second Edition, ACM Press Books
2. C. Manning, P. Raghavan, and H. Schütze, ―Introduction to Information Retrieval, Cambridge University Press

Additional References:

1. Ricci, F, Rokach, L. Shapira, B. Kantor, ―Recommender Systems Handbook‖, First Edition.
2. Bruce Croft, Donald Metzler, and Trevor Strohman, Search Engines: Information Retrieval in Practice, Pearson Education.
3. Stefan Buttcher, Charlie Clarke, Gordon Cormack, Information Retrieval: Implementing and Evaluating Search Engines, MIT Press.

Course/ Paper Title	**Linux Server Administration**
Course offered as	**Discipline Specific Elective - II**
Course Code	**RUCSDSE5042**
Semester	**V**
No. of Credits	**2 (30 hours)**
No. of lecture Hours/week	**2**

Sr No.	Course Learning Objectives:
CLO1	Explain Linux fundamentals including distributions, file hierarchy, and essential command usage.
CLO2	Develop understanding of Linux server administration principles and concepts.
CLO3	Acquire practical skills in managing users, groups, and file systems in a Linux environment.
CLO4	To configure and secure network services such as DNS, FTP, SMTP, SSH.
CLO5	Gain knowledge of advanced network administration topics including NFS, Samba, DFS, NIS, LDAP, DHCP applications, file services, email services, and VPN

Course Outcome

	On completing the course, the student will be able to:
CO1	Use Linux commands and navigate the Linux File Hierarchy Standard to perform basic system operations.
CO2	Create and manage users and groups, control boot/shutdown, and manage services using systemd/systemctl.
CO3	Install and configure DNS, FTP, web server, mail services (SMTP, POP, IMAP), SSH, and Kerberos-based authentication.
CO4	Configure TCP/IP networking and implement firewall rules using Netfilter for system security.
CO5	Configure and manage NFS, Samba, DFS, NIS, LDAP, DHCP, VPN, and related network applications.

Detailed Syllabus

Module	Title with content	No. of lectures
I	**Introduction:** Introduction of Linux operating system, Linux Distributions, Linux file hierarchy (FHS) , Basic Linux commands. **Single-Host Administration**: Managing Users and Groups, Booting and shutting down processes. Core System Services using systemd and systemctl, Process of configuring, compiling, Linux Kernel. **Networking and Security**: TCP/IP for System Administrators, basic network Configuration, Linux Firewall (Netfilter), System and network security	**15**
II	**Internet Services**: Domain Name System (DNS), File Transfer Protocol (FTP), Apache web server, Simple Mail Transfer Protocol (SMPT), Post Office Protocol and Internet Mail Access Protocol (POP and IMAP), Secure Shell (SSH), Network authentication system (Kerberos commands), Domain Name Service (DNS), Security. **Internet Services**: Network File System (NFS), Samba, Distributed File Systems (DFS),Network Information Service (NIS), Lightweight Directory Access Protocol (LDAP), Dynamic Host configuration Protocol (DHCP), Applications, NFS, Email Services, VPN.	**15**

References:

1. Linux Administration: A Beginner's Guide, Wale Soyinka, Seventh Edition, McGraw-Hill Education, 2016 2. Ubuntu Server Guide, Ubuntu Documentation Team, 2016
2. UNIX and Linux System Administration Handbook — Evi Nemeth, Garth Snyder, Trent R. Hein, Ben Whaley (Pearson)

Additional References:

1. Mastering Ubuntu Server, Jay LaCroix, PACKT Publisher, 2016
2. Linux Pocket Guide — Daniel J. Barrett (O'Reilly) ,3rd Edition , 2016

Course/ Paper Title	**Cloud Computing & Web Services**
Course offered as	**Minor**
Course Code	**RUCSMN505**
Semester	**V**
No. of Credits	**2 (30 Hrs)**
No. of lecture Hours/week	**2**

Sr No.	Course Learning Objectives:
CLO1	Understand and design web service technologies using SOAP, REST, WSDL, JAX-WS, and JAX-RS.
CLO2	Explain the basic concepts of parallel computing and distributed computing to execute programs efficiently using concurrent processing principles.
CLO3	Explain and apply virtualization concepts by creating and managing virtual machines using KVM and administering virtualized environments through oVirt.
CLO4	Understand the basics of cloud computing and essential characteristics of cloud platforms
CLO5	Apply cloud tools like CloudSim and OpenStack for simulation and cloud operations.
CLO6	Analyze cloud service models and design cloud network solutions using Amazon Web Services VPC.

Course Outcome

	On completing the course, the student will be able to:
CO1	Apply web service technologies by designing and implementing SOAP and RESTful services using WSDL, JAX-WS, and JAX-RS.
CO2	Demonstrate virtualization by creating and managing virtual machines using KVM and managing environments through oVirt.
CO3	Explain fundamental cloud computing concepts, deployment models, and security practices.
CO4	Apply cloud technologies using tools such as CloudSim and OpenStack to simulate and manage cloud environments.
CO5	Analyze cloud service models and compare cloud providers also design cloud networking solutions using Amazon Web Services Virtual Private Cloud (VPC).

Detailed Syllabus

Module	Title with content	No. of lectures
I	**Web Services and Distributed Applications –** Introduction to Web Services, Features, types of web services, Basics of Distributed and Parallel Computing. **SOAP & RESTful Web Services:** Introduction of SOAP, Advantages, Structure of SOAP Message, SOAP Messaging Architecture, SOAP Header and faults, Introduction of RESTful Web Services, REST architecture Constraints, REST API Design Principles, HTTP methods, Java API for RESTful Web Services (JAXRS), Comparison of SOAP vs REST **Virtulization:** - Concept of Virtulization, Advantages and disadvantages Virtualization, Types of Virtualizations, Hypervisors, role of Virtualization in Cloud Computing, Introduction of KVM, creating virtual machines using KVM, oVirt - management tool for virtualization environment.	**15**
II	**Introduction to Cloud Computing:** Definition, Types of Clouds, Deployment of software solutions and web applications, Comparing cloud providers with traditional IT service providers, Cloud Information Security Objectives, Secure Development practices. **Cloud Applications:** CloudSim: Introduction to Simulator, understanding CloudSim simulator, CloudSim Architecture. **OpenStack**: Introduction to OpenStack, Basic OpenStack operations, OpenStack CLI and APIs, Tenant model operations, Quotas, Private cloud building blocks. **AWS**: Architecting on AWS, Building complex solutions with Amazon Virtual Private Cloud (Amazon VPC)	**15**

References:

1. Pro Power Shell for Amazon Web Services, Brian Beach, Apress, 2014
2. Enterprise Cloud Computing Technology, Architecture, Applications, Gautam Shroff, Cambridge University Press, 2010
3. OpenStack in Action, V. K. CODY BUMGARDNER, Manning Publications Co, 2016

Additional References:

1. OpenStack Operations Guide, Tom Fifield, Diane Fleming, Anne Gentle, Lorin Hochstein, Jonathan Proulx, Everett Toews, and Joe Topjian, O'Reilly Media, Inc., 2014
2. Mastering Cloud Computing, Rajkumar Buyya, Christian Vecchiola, S Thamarai Selvi, Tata McGraw Hill Education Private Limited, 2013

Course/ Paper Title	**Cloud Computing & Web Services- Practical**
Course offered as	**Minor**
Course Code	**RUCSMNP505**
Semester	**V**
No. of Credits	**2 (60 Hrs)**
No. of lecture Hours/week	**2**

Sr No.	**Course Learning Objectives:**
CLO 1	Develop and deploy basic SOAP and RESTful web services to perform real-world operations.
CLO 2	Design and implement interoperable web services and applications by integrating services across platforms and consuming external APIs.
CLO 3	Apply cloud computing concepts by configuring virtualization environments, implementing IaaS/PaaS services, and managing cloud resources.
CLO 4	Demonstrate file transfer techniques using MTOM and configure basic cloud security.
CLO 5	Apply cloud security and data transfer techniques in practical applications.

Course Outcome

	On completing the course, the student will be able to:
CO1	Create and deploy SOAP and RESTful web services for real-world tasks.
CO2	Develop and integrate web services across platforms (Java and .NET) and consume external APIs.
CO3	Implement cloud infrastructure using virtualization, OpenStack, and IaaS/PaaS services.
CO4	Apply file transfer techniques (MTOM) and configure basic cloud security mechanisms.
CO5	Analyze and compare major cloud platforms such as AWS, Azure, and Google Cloud.

Sr No.	Experiments
1	Create a Simple SOAP service to convert temperature from Fahrenheit to Celsius and vice versa.
2	Write a SOAP web service program to implement the operation can receive request and will return a response in two ways. a) One - Way operation b) Request –Response
3	Create a SOAP web service that performs basic calculator operations (add, subtract, multiply, divide).
4	Create a Simple REST Service to store employee details like employee id,name and designation. Perform CRUD Operations,
5	Create REST API that returns data in both JSON and XML formats.
6	Write a program to implement business UDDI Registry entry.
7	Define a simple services like Converting Rs into Dollar and Call it from different platform like JAVA and .NET
8	Develop an application to consume Google's search / Google's Map RESTful Web service.
9	Develop application to download image/video from server or upload image/video to server using MTOM techniques
10	Installation and Configuration of virtualization using KVM.
11	Implement FOSS-Cloud Functionality VSI (Virtual Server Infrastructure) Infrastructure as a Service (IaaS).
12	Create users, instances, and private networks in OpenStack.
13	Case study of major cloud platforms: Amazon Web Services, Microsoft Azure, and Google Cloud Platform (using documentation/free tiers).
14	Study and configure basic security mechanisms such as authentication, firewall rules, and secure access in cloud environments.
15	Study and implementation of Single-Sign-On (SSO).
16	Create an account on Amazon Web Services and explore the dashboard.
17	Study and implementation of Platform as a Service.
18	Study and implementation of Software as a Service.
19	Study and implementation of Storage as a Service.
20	To implement secure user management in a cloud environment by creating users, groups, roles, and policies, and testing controlled access to resources.

Course/ Paper Title	**Ethical Hacking & Cyber Forensics**
Course offered as	**VSC**
Course Code	**RUCSVSC506**
Semester	**V**
No. of Credits	**2 (60 Hrs)**
No. of lecture Hours/week	**4**

Sr No.	**Course Learning Objectives:**
CLO1	To introduce students to cybersecurity fundamentals including reconnaissance, footprinting, and vulnerability assessment techniques used in ethical hacking.
CLO2	To provide hands-on experience with cryptographic concepts by implementing encryption and decryption techniques using tools such as CrypTool.
CLO3	To understand malware behavior and system monitoring techniques, denial-of-service (DoS) attack mechanisms.
CLO4	Create and verify forensic images using tools such as FTK Imager and EnCase Imager, and validate data integrity using hash verification techniques.
CLO5	Perform secure data acquisition, Recover and analyze deleted files using forensic tools and manual command-line techniques.
CLO6	Perform mobile device forensics to extract and analyze call logs, messages, Conduct email forensic analysis for identifying spoofing or tampering and tracing the origin of suspicious communications. Analyze web browser artifacts to reconstruct user activity and internet usage patterns.

Course Outcome

	On completing the course, the student will be able to:
CO1	Apply cryptographic techniques to encrypt and decrypt data using symmetric key algorithms and validate the correctness of the encryption process.
CO2	Demonstrate understanding of malware behavior by developing and analyzing a basic keylogging mechanism.
CO3	Identify and exploit common web application vulnerabilities such as persistent Cross-Site Scripting (XSS) and SQL Injection.Evaluate the impact of Denial-of-Service (DoS) attacks.
CO4	Recover, inspect, and analyze deleted or hidden files using forensic software and command-line techniques.
CO5	Perform mobile device forensic analysis to extract and interpret call logs, messages, and other relevant artifacts. Analyze email evidence by examining headers and identifying spoofing, tampering, or malicious intent**.**
CO6	Investigate USB device artifacts to identify previously connected removable storage devices and related system traces.

Sr No.	Ethical Hacking & Cyber Foresencis
1	Google and Whois Reconnaissance • Use Google search techniques to gather information about a specific target or organization. • Utilize advanced search operators to refine search results and access hidden information. • Perform Whois lookups to retrieve domain registration information and gather details about the target's infrastructure.
2	Password Encryption and Decryption: • Use CrypTool to encrypt passwords using the RC4 algorithm. • Decrypt the encrypted passwords and verify the original values.
3	Creating a Keylogger with Python • Write a Python script that captures and logs keystrokes from a target system. • Execute the keylogger script and observe the logged keystrokes.
4	Run and analyze the output of following commands in Linux – • ifconfig, • ping, • netstat, • traceroute
5	Use NMap scanner to perform port scanning of various forms – ACK, SYN, FIN, NULL, XMAS.
6	Simulate persistent cross-site scripting attack. • Set up a vulnerable web application that is susceptible to persistent XSS attacks. • Craft a malicious script to exploit the XSS vulnerability and execute arbitrary code.
	• Observe the consequences of the attack and understand the potential risks associated with XSS vulnerabilities.
7	Use Nemesy to launch a DoS attack against a target system or network. Observe the impact of the attack on the target's availability and performance.
8	Session Impersonation with Firefox and Tamper Data. • Install and configure the Tamper Data add-on in Firefox. • Intercept and modify HTTP requests to impersonate a user's session.
9	Perform SQL injection attack. • Identify a web application vulnerable to SQL injection. • Craft and execute SQL injection queries to exploit the vulnerability.
10	Exploiting with Metasploit (Kali Linux). • Identify a vulnerable system and exploit it using Metasploit modules. • Gain unauthorized access to the target system and execute commands or extract information. • Understand the ethical considerations and legal implications of using Metasploit for penetration testing.

11	Creating a Forensic Image using FTK Imager/Encase Imager : • Creating Forensic Image • Check Integrity of Data • Analyze Forensic Image
12	Data Acquisition: • Perform data acquisition using: • USB Write Blocker + Encase Imager • SATA Write Blocker + Encase Imager • Falcon Imaging Device
13	Recovering and Inspecting deleted files • Check for Deleted Files • Recover the Deleted Files • Analyzing and Inspecting the recovered files • Perform this using recovery option in ENCASE and also Perform manually through command line
14	Forensics Case Study: Solve the Case study (image file) provide in lab using Encase Investigator or Autopsy.
15	Using Sysinternals tools for Network Tracking and Process Monitoring : • Check Sysinternals tools • Monitor Live Processes • Capture RAM • Capture TCP/UDP packets • Monitor Hard Disk • Monitor Virtual Memory • Monitor Cache Memory
16	Mobile Device Forensics • Perform a forensic analysis of a mobile device, such as a smartphone or tablet.
	• Retrieve call logs, text messages, and other relevant data for investigative purposes.
17	Email Forensics • Analyze email headers and content to trace the origin of suspicious emails. • Identify potential email forgeries or tampering.
18	Web Browser Forensics • Analyze browser artifacts, including history files, bookmarks, and download records. • Analyze cache and cookies data to reconstruct user-browsing history and identify visited websites or online activities. • Extract the relevant log or timestamp file, analyze its contents and interpret the timestamp data to determine the user's last internet activity and associated details.
19	To study the steps for hiding and extract any text file behind an image file/ Audio file using Command Prompt.
20	To find last connected USB on your system (USB Forensics)

SEMESTER VI

Course/ Paper Title	**Cryptography and Network Security**
Course offered as	**Major I**
Course Code	**RUCSMJ601**
Semester	**VI**
No. of Credits	**4 (60 Hrs)**
No. of lecture Hours/week	**4**

Sr No.	Course Learning Objectives:
1	Analyze security attacks, trends, and protection mechanisms using the OSI security architecture to propose appropriate security solutions
2	Apply classical encryption methods such as substitution and transposition ciphers to encrypt/decrypt data and evaluate their effectiveness.
3	Implement public-key cryptographic algorithms such as RSA for secure key exchange and communication.
4	Design secure authentication systems using message authentication codes and digital signatures to ensure integrity and non-repudiation.
5	Evaluate and configure network security mechanisms such as IP security, Transport Layer Security, intrusion detection systems, and firewall rules to protect systems from attacks.

Course Outcome

Detailed Syllabus

	On completing the course, the student will be able to:
CO1	Explain the fundamental principles, models, and mechanisms of computer and network security based on the OSI security architecture.
CO2	Demonstrate classical, symmetric, and public-key encryption techniques and differentiate their strengths and limitations.
CO3	Illustrate authentication, key management, message authentication, and digital signature mechanisms used in secure communication.
CO4	Analyze network security techniques including IP security, web security, intrusion detection, malware threats, and firewall design.
CO5	Apply security practices to design secure communication systems and protect networks from common attacks.

Module	Topics	No. of Lectures
I	**Introduction and Classical Cryptography**: Introduction to cryptography and network security, Security concepts (CIA), Security Trends. OSI Security Architecture: Security Attacks, Security Services, Security Mechanisms. **Classical Encryption Techniques:** Symmetric Cipher Model, Substitution Techniques (Caesar, Monoalphabetic, Vigenère, Vernam, Playfair) and Transposition ciphers (Rail-fence, Columnar), Steganography.	**15**

II	**Block and Stream Cipher Principles:** Shannon's theory of confusion and diffusion, Feistel network structure. , The Data Encryption Standard (DES), Advanced Encryption Standard (AES), Multiple Encryption, Differentiate between DES and AES. Block Cipher Modes of Operation, Stream Ciphers, RC4. **Public-Key Cryptography and RSA:** Principles of Public-Key Cryptosystems, The RSA Algorithm.	**15**
III	**Key Management:** Public-Key Cryptosystems, Key Management, Diffie Hellman Key Exchange. **Data Integrity in Cryptography**: Threats to Data Integrity, Message Authentication and Hash Functions, Security of Hash Functions and Macs, Secure Hash Algorithm, HMAC, Digital Signatures, Digital Signature Standard. **Authentication Applications:** Kerberos, X.509 Authentication, Public-Key Infrastructure	**15**
IV	**Web Security:** Secure Socket Layer (SSL) and Transport Layer Security (TLS). **Email Security**: Pretty Good Privacy (PGP), S/MIME. **IP Security**: IP Security Overview (IPSec), Authentication Header (AH), Encapsulating Security Payload (ESP). **Intrusion**: Intruders, Intrusion Techniques, Intrusion Detection **System Security:** Firewalls and its types, Viruses and Related Threats, Virus Countermeasures, DDOS.	**15**

References:

1. Cryptography and Network Security: Principles and Practice 7th edition, William Stallings, Pearson

Additional Reference(s):

1. Cryptography and Network, 2nd edition, Behrouz A Fourouzan, Debdeep Mukhopadhyay, TMH.

2. Cryptography and Network Security, 4th Edition , Atul Kahate , McGraw-Hill Education

Course/ Paper Title	**Software Testing & Quality Assurance**
Course offered as	**Major II**
Course Code	**RUCSMJ602**
Semester	**VI**
No. of Credits	**4 (60 Hrs.)**
No. of lecture Hours/week	**4**

Sr No.	**Course Learning Objectives:**
CLO 1	Explain software quality concepts, SQA activities, reviews, FTR and Software Configuration Management.
CLO 2	Apply verification and validation principles, testing fundamentals and test case design techniques to develop test plans and documentation.
CLO 3	Design and implement white-box, black-box, and experience-based testing strategies across unit, integration, validation, and system testing levels.
CLO 4	Analyse and evaluate software quality using software metrics, reliability measures, complexity metrics, and statistical process control techniques.
CLO 5	Manage defect life cycle processes and apply ISO standards, Six Sigma and quality improvement tools for continuous process improvement.

Course Outcome

	On completing the course, the student will be able to:
CO1	Describe software quality concepts, SQA activities, reviews, SCM processes.
CO2	Develop test plans and design effective test cases using verification, validation, white-box, and black-box testing techniques.
CO3	Apply appropriate testing strategies at unit, integration, validation, and system testing levels.
CO4	Evaluate software quality using software metrics, reliability measures, and statistical quality control methods.
CO5	Manage defects and implement quality improvement techniques using standard quality models and tools.

Detailed Syllabus

Module	Topics	No. of Lectures
I	**Introduction to Software Testing:** Nature of errors and the need for testing **Definition of Quality and Quality Assurance:** Understanding quality in software development, Distinction between Quality Assurance (QA), Quality Control (QC), Quality Management (QM). **Software Quality Assurance:** Understanding quality concepts and the Quality Movement: Background issues and challenges in SQA, SQA activities, Software reviews, FTR, SQA plan **Software Configuration Management:** elements of SCM, SCM Process, Change Control Capability Maturity Model.	**15**
II	**Software Testing :** Verification and Validation, Introduction to Testing, Testing Principles, Testing Objectives, Test Oracles, **Testing Fundamentals:** Basics of software testing process, Test case design principles and techniques, Test execution, reporting, and documentation **Testing Types:** Levels of Testing, White-Box Testing/Structural Testing, Functional/Black-Box Testing, Test Plan, Test-Case Design.	**15**
III	**Software Testing Techniques and Strategies :** White Box Testing and Black Box Testing: Functional/Specification based Testing as Black Box, Structural Testing as White Box, Experience-based. **Software Testing Strategies:** Strategic approach to software testing Unit Testing, Integration Testing, Validation Testing, System Testing. **Software Metrics:** Concept of software metrics and their importance, Software reliability and measures, Developing and utilizing different types of metrics, Complexity metrics and their significance in testing	**15**
IV	**Defect Management:** Definition of defects and their lifecycle, Defect management process, including defect reporting and tracking. **Statistical process control techniques for quality assurance:** Software reliability measurement and improvement, The ISO 9000 Quality Standards, Six sigma and their application in software development **Quality Improvement Techniques:** Introduction to quality improvement methodologies, Utilizing quality costs for decision-making, Introduction to quality improvement tools: Pareto Diagrams, Cause-effect Diagrams, Scatter Diagrams, Run charts	**15**

References:

1. Software Engineering for Students, A Programming Approach, Douglas Bell, 4th Edition, Pearson Education, 2005
2. Software Engineering – A Practitioners Approach, Roger S. Pressman, 7th Edition, Tata McGraw Hill

Additional Reference(s):

1. Quality Management, Donna C. S. Summers, 5th Edition, Prentice-Hall.
2. Software Testing and Quality Assurance Theory and Practice, Kshirsagar Naik, Priyadarshi Tripathy , John Wiley & Sons, Inc. , Publication.

Course/ Paper Title	**Computer Science Practical 6**
Course offered as	**Major Practical**
Course Code	**RUCSMJP6**
Semester	**V**
No. of Credits	2 (60 Hrs.)
No. of lecture Hours/week	2

Sr. No.	**Course Learning Objectives of Module I & II**
CLO1	To understand and apply encryption and decryption using substitution and transposition techniques with suitable tools/libraries.
CLO2	To implement public-key cryptography algorithms such as RSA for secure key exchange and communication.
CLO3	Design and implement secure authentication mechanisms using message authentication codes and digital signatures.
CLO4	Understand the concept of RC4 Stream Cipher and implement it.
CLO5	Evaluate and implement various security measures, such as IP security, web security protocols and firewall configurations, to protect networks and systems from unauthorized access and attacks.
CLO6	Apply Selenium IDE to create and execute test suites for web applications.
CLO7	Demonstrate Selenium Server and WebDriver for automating browser tasks and login
CLO8	Analyze and manipulate web and Excel data using automation scripts.
CLO9	Evaluate web elements by identifying and counting components on web pages.
CLO10	Implement load testing using JMeter to assess web application performance.

Course Outcome

	On completing the course, the student will be able to:
CO1	Apply substitution and transposition techniques using tools/libraries to perform encryption and decryption of sample data.
CO2	Implement public-key cryptography algorithms such as RSA to securely exchange keys and establish secure communication.
CO3	Design and implement authentication mechanisms using message authentication codes and digital signatures.
CO4	Explain the concept of RC4 stream cipher and implement its encryption and decryption process.
CO5	Evaluate and implement security measures such as IP security, web security protocols, and firewall rules to protect networks and systems.
CO6	Apply Selenium IDE to design and execute automated test cases for web applications.
CO7	Demonstrate the use of Selenium WebDriver and Selenium Server for automating web interactions and validations.
CO8	Analyze and process data from web pages and external files (e.g., Excel) using automation tools.
CO9	Evaluate web application elements by identifying, verifying, and counting UI components.
CO10	Implement performance testing using JMeter to assess and interpret system behavior under load.

Sr No.	**Module I : Cryptography and Network Security Practical**
1	Implementing Substitution Ciphers: Design and implement algorithms to encrypt and decrypt messages using classical substitution and transposition techniques a) Caser Cipher b) monoalphabetic
2	Implementing Transposition Ciphers: Design and implement algorithms to encrypt and decrypt messages using classical substitution and transposition techniques a) Rail Fence 2) Columnar Transposition
3	To encrypt and decrypt a message using the Caesar Cipher with CrypTool 2.
4	Comprehensive Study and Implementation of RC4 Stream Cipher
5	Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.
6	Implement algorithms to generate and verify message authentication codes (MACs) for ensuring data integrity and authenticity.

7	Implement digital signature algorithms such as RSA-based signatures, and verify the integrity and authenticity of digitally signed messages.
8	Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.
9	Configure IPsec on network devices to provide secure communication and protect against unauthorized access and attacks.
10	Analyze and identify malware samples using antivirus tools, analyze their behavior, and develop countermeasures to mitigate their impact.
11	Configure and test firewall rules to control network traffic, filter packets based on specified criteria, and protect network resources from unauthorized access.
12	Use Wireshark to capture network traffic and analyze it.
	Note: Any ten practicals have to be performed.
Sr No.	**Module II: Software Testing & Quality Assurance Practical**
1	Install Selenium IDE and create a test suite containing a minimum of 4 test cases for different web page formats (e.g., HTML, XML, JSON, etc.).
2	Conduct a test suite for two different websites using Selenium IDE. Perform various actions like clicking links, filling forms, and verifying content.
3	Install Selenium Server (Selenium RC) and demonstrate its usage by executing a script in Java or PHP to automate browser actions.

4	Write a program using Selenium WebDriver to automate the login process on a specific web page. Verify successful login with appropriate assertions.
5	Write a program using Selenium WebDriver to update 10 student records in an Excel file. Perform data manipulation and verification.
6	Write a program using Selenium WebDriver to select the number of students who have scored more than 60 in any one subject (or all subjects). Perform data extraction and analysis.
7	Write a program using Selenium WebDriver to provide the total number of objects present or available on a web page. Perform object identification and counting.
8	Write a program using Selenium WebDriver to get the number of items in a list or combo box on a web page. Perform element identification and counting.
9	Write a program using Selenium WebDriver to count the number of checkboxes on a web page, including checked and unchecked counts. Perform checkbox identification and counting.
10	Perform load testing on a web application using JMeter.

Course/ Paper Title	**Game Programming**
Course offered as	**Discipline Specific Elective - I**
Course Code	**RUCSDSE6031**
Semester	**VI**
No. of Credits	2 (30 Hrs)
No. of lecture Hours/week	2

Sr No.	Course Learning Objectives:
CLO1	To understand the architecture and design of modern game engines.
CLO2	To understand resource management, game loop control, collision handling, and user interface management in games.
CLO3	To familiarize students with game development environments such as Unity and DX Studio.
CLO4	To explain the fundamentals of DirectX 11 including textures, buffers, swap chain, and rendering concepts.
CLO5	To develop knowledge of graphics programming using DirectX 11.
CLO6	To design and implement 2D and 3D games using Python frameworks.
CLO 7	Develop skills in managing game objects, classes, functions and design advanced game features including animation, sound, UI, multiple scenes, AI, and publishing.

Course Outcome

	On completing the course, the student will be able to:
CO1	Explain game engine architecture, support systems, resource handling, and real-time simulation concepts.
CO2	Apply game loop control, collision detection, memory management, UI management, and event handling in game development.
CO3	Explain DirectX 11 concepts such as textures, resource formats, swap chain, depth buffering, and multisampling.
CO4	Develop 2D and basic 3D games using Python frameworks, Implement core engine components.
CO5	Demonstrate understanding of DirectX 11 fundamentals.
CO6	Apply programming constructs such as decision control, loops, arrays, functions, classes, exception handling, and debugging in game development.
CO7	Use the Unity development environment effectively to create interactive games.

Detailed Syllabus

Module	Title with content	No. of lectures
I	**Game Engines and Design:** Game engine architecture, Engine support systems, Resources and File systems, Game loop and real-time simulation, Human Interface devices, Collision and rigid body dynamics, Game profiling. Introduction to Dx Studio, Introduction to Unity, Application layer, Game logic, Game views, managing memory, controlling the main loop, loading and caching game data, User Interface management, Game event management. **Introduction To DirectX 11:** COM, Textures and Resources Formats, The swap chain and Page flipping, Depth Buffering, Texture Resource Views, Multisampling Theory and MS in Direct3D, Feature Levels **Game Platform and Development**: 2D and 3D Game development using Python-Pygame, ModernGL, Numpy, PyGLM,Ursina (OpenGL)	**15**
II	**Unity Development Environment:** IDE Basics, Unity Concepts, Sprites, Game Loops and Functions, Game Design Strategies. **Simple Movement, Operations and Object Oriented Concepts:** Simple Rotation and Scaling, Rigidbody Components, Unity Colliders, Physics Materials, Scripting Collision Events, Primitive Data and Math, Decisions and Flow Control, Loops and Arrays, Game Design Strategies, Exceptions and Debugging, Defining Classes, Functions, Organizing and Managing Game Objects, **Advanced Programming:** Virtual Worlds, Scrolling Games, Animation, Sound Effects, Advanced Game Physics, Multiple Scenes, Artificial Intelligence, User Interfaces, Game Art, Publishing Games.	**15**

References:

1. Introduction To 3D Game Programming With Directx® 11,Frank D Luna, Mercury Learning And Information,2012.
2. Unity in Action, Third Edition ,Joe Hocking,Manning Publications, 2024.
3. Game Engine Architecture,Jason Gregory,Edition: 3rd,A K Peters/CRC Press,2018.
4. Beginning Python Games Development, Second Edition,Harrison Kinsley & Will McGugan,Apress (Springer),2015.

Additional References:

1. Python, PyGame, and Raspberry Pi Game Development,Sloan Kelly,Apress,2019.
2. Game Coding Complete, Fourth Edition,Mike McShaffry, David Graham,CRC Press,2014.
3. Unity Game Development Cookbook,Paris Buttfield-Addison et al,O'Reilly,2017.

Course/ Paper Title	**Wireless Sensor Network**
Course offered as	**Discipline Specific Elective – II**
Course Code	**RUCSDSE6032**
Semester	**VI**
No. of Credits	2 (30 Hrs)
No. of lecture Hours/week	2

Sr. No.	**Course Learning Objectives:**
CLO1	Explain the fundamentals, characteristics, architecture, and applications of Wireless Sensor Networks (WSNs).
CLO 2	Describe sensor node technology, radio technology, operating environment, and network architecture in Wireless Sensor Networks.
CLO 3	Explain and compare Wireless Sensor Network operating systems and their role in resource-constrained environments.
CLO 4	Discuss the characteristics, challenges, energy efficiency, and security and privacy issues of ad-hoc networks in WSNs.
CLO 5	To introduce the fundamentals of MAC protocols and routing mechanisms used in Wireless Sensor Networks.
CLO 6	To understand TCP design issues and middleware architecture in WSN environments.
CLO 7	To explain wireless transmission concepts such as signals, antennas, modulation, multiplexing, and spread spectrum.
CLO 8	To study cellular, telecommunication, and satellite communication systems including GSM, DECT.

Course Outcome

	On completing the course, the student will be able to:
CO 1	Define Wireless Sensor Networks and list their characteristics, advantages, challenges, and applications.
CO 2	Illustrate the basic architectural elements, sensor node components, radio technology, and operating environment of WSNs.
CO 3	Explain optimization goals, design principles, service interfaces, and gateway concepts in WSNs.
CO 4	Explain the concept of ad-hoc networks in WSNs with respect to energy efficiency, security, and privacy challenges.
CO 5	Explain routing challenges, design issues, and routing strategies used in Wireless Sensor Networks.
CO 6	Describe the fundamentals of Medium Access Control protocols and the working of Sensor-MAC in Wireless Sensor Networks.
CO 7	Describe TCP design issues and middleware architecture in WSN.
CO 8	Explain the architecture, protocols, and services of cellular and telecommunication systems such as GSM, DECT.

Detailed Syllabus

Module	Topics	No. of Lectures
I	**Introduction and Overview of Wireless Sensor Networks: Introduction to wireless sensor network and its characteristics,** Basic Sensor Network Architectural Elements, Advantage and challenges, Applications, Sensor Node Technology, , WN Operating Environment, Radio Technology, Network architecture, Optimization goals and figures of merit, Design principles for WSNs, Service interfaces of WSNs, Gateway concepts. **Wireless Sensor Network Operating Systems and Ad-hoc Networks:** Overview of Wireless Sensor Network Operating Systems, Examples of WSN Operating Systems Ad-hoc Networks in Wireless Sensor Networks, Characteristics and Challenges of Ad-hoc Networks in WSNs, Energy Efficiency Considerations in Ad-hoc Networks, Security and Privacy in Adhoc Networks, Examples of WSN OS, Ad-hoc Network.	**15**
II	**Medium Access Control Protocol**: Fundamentals of MAC Protocols, Sensor-MAC Case Study **Routing in WSN:** Routing Challenges and Design Issues in Wireless Sensor Networks, Routing Strategies in Wireless Sensor Networks **Transport Control Protocol:** Introduction of TCP, TCP Design Issues, WSN Middleware Architecture **Wireless Transmission:** Frequency for radio transmission, Signals, Antennas, Signal propagation, Multiplexing, Modulation, Spread spectrum, Cellular systems. **Telecommunication, Satellite and Broadcast Systems:** Satellite and Broadcast Systems: GSM: Mobile services, System architecture, Radio interface, Protocols, Localization And Calling, Handover, security, New data services; DECT: System architecture, Protocol architecture, Compare GSM and DECT.	**15**

References:

1. Wireless Sensor Networks Technology, Protocols, and Applications ,Kazem Sohraby, Daniel Minoli and TaiebZnati, John Wiley & Sons, 2017
2. Protocols and Architectures for Wireless Sensor Network, Holger Kerl, Andreas Willig, John Wiley and Sons, 2015

Additional Reference(s):

1. Fundamentals of Wireless Sensor Networks, Theory and Practice, Waltenegus Dargie, Christian Poellabauer , Wiley Series on wireless Communication and Mobile Computing, 2011
2. Networking Wireless Sensors, Bhaskar Krishnamachari , Cambridge University Press, 2005

Course/ Paper Title	**Data Mining & Warehousing**
Course offered as	**Minor**
Course Code	**RUCSMN604**
Semester	**VI**
No. of Credits	2 (30 Hrs.)
No. of lecture Hours/week	2

Sr No.	**Course Learning Objectives:**
CLO1	Describe and explain the fundamental concepts of Data Warehousing, architecture, dimensional modelling, ETL processes and OLAP systems.
CLO2	Explain OLTP, OLAP systems and multidimensional data operations.
CLO3	Apply data exploration and preprocessing techniques to organize and prepare data for analytical and mining tasks.
CLO4	To introduce the fundamentals, architecture, tasks, and applications of Data Mining.
CLO5	Analyse large datasets using association rule mining and frequent pattern mining techniques to discover meaningful patterns.
CLO6	Evaluate and design appropriate data mining solutions for real-world problems including web, sequence and time-series data analysis.

Course Outcome

	On completing the course, the student will be able to:
CO1	Explain the concepts of Data Warehousing, dimensional modeling, ETL process.
CO2	Compare OLTP and OLAP systems and perform multidimensional OLAP operations such as slice, dice, roll-up, drill-down, and pivot.
CO3	Apply data exploration and preprocessing techniques to prepare data for mining tasks.
CO4	Explain data mining concepts, architecture, task primitives, issues, metrics, and applications.
CO5	Analyze datasets using frequent pattern mining and association rule algorithms such as Apriori and FP-Growth.
CO6	Evaluate and select suitable data mining techniques for solving real-world analytical problems including web and time-series data.

Detailed Syllabus

Module	Title with content	No. of lectures
I	**Introduction and Architecture of Data Warehouse** Introduction to Data Warehouse, Necessity and Objectives of Data Warehousing, Framework of Data Warehouse, Components of Data Warehouse System, Data Warehouse Architecture (Three-tier Architecture), Metadata Repository, Developing Data Warehouses, Data Warehouse versus Data Mart. **Data Warehouse Models and Dimensional Design** Enterprise Data Warehouse (EDW), Data Mart, Virtual Data Warehouse, Hybrid Data Warehouse, E-R Modelling versus Dimensional Modelling, Defining Dimensional Model, Information Package Diagram, Granularity of Facts, Additivity of Facts, Updates to Dimension Tables. **Data Warehouse Schemas and ETL Process** Star Schema, Snowflake Schema, Fact Constellation Schema, Major Steps in ETL Process (Extraction, Transformation, Loading), Data Extraction Strategies, Data Loading Techniques. **OLTP, OLAP and Multidimensional Operations** OLTP (Online Transaction Processing), OLAP (Online Analytical Processing), Differences between OLTP and OLAP, Types of OLAP Servers (ROLAP, MOLAP, HOLAP), OLAP Operations in Multidimensional Data Model including Slice, Dice, Roll-up, Drill-down and Pivot.	**15**
II	**Introduction to Data Mining and Architecture** Introduction to Data Mining, Definition of Data Mining, Data Mining Architecture, Data Mining Task Primitives, Kinds of Databases, Data to be Mined, Basic Data Mining Techniques, Data Mining Issues, Data Mining Metrics, Social Implications of Data Mining, Applications of Data Mining. **Data Exploration and Preprocessing for Mining** Data Exploration, Types of Attributes, Statistical Description of Data, Data Visualization, Descriptive Data Summarization, Data Cleaning, Data Integration, Data Transformation, Data Reduction, Data Discretization, Concept Hierarchy Generation. **Mining Frequent Patterns and Association Rules** Market Basket Analysis, Frequent Itemset, Closed Frequent Itemset, Maximal Frequent Itemset, Association Rules, Apriori Algorithm, Improving Efficiency of Apriori, Mining Frequent Itemset without Candidate Generation, FP-Growth Algorithm, Mining Multilevel Association Rules, Mining Multidimensional Association Rules. **Advanced Mining Techniques and Web Mining** Web Mining, Web Content Mining including Crawlers, Harvest System, Virtual Web View and Personalization, Web Structure Mining including PageRank and CLEVER Algorithm, Web Usage Mining, Mining Time-Series Data, Mining Sequence Data, Mining Text Databases, Mining the World Wide Web.	**15**

References:

1. J. Han, M. Kamber, and J. Pei, *Data Mining: Concepts and Techniques*, 3rd ed. Burlington, MA, USA: Morgan Kaufmann, 2011.
2. Berson and S. J. Smith, *Data Warehousing, Data Mining, and OLAP*, 2nd ed. New York, NY, USA: McGraw-Hill, 2002.

Additional References:

1. P. Ponniah, *Data Warehousing: Fundamentals for IT Professionals*, 2nd ed. Hoboken, NJ, USA: Wiley, 2010.
2. P.-N. Tan, M. Steinbach, and V. Kumar, *Introduction to Data Mining*, 2nd ed. Boston, MA, USA: Pearson, 2019.

Course/ Paper Title	**Data Mining & Warehousing- Practical**
Course offered as	**Minor Practical**
Course Code	**RUCSMNP604**
Semester	**VI**
No. of Credits	2 (60 Hrs.)
No. of lecture Hours/week	4

Sr. No.	Course Learning Objectives
CLO1	Understand the concepts of data warehousing, dimensional modeling, ETL processes, and OLAP operations.
CLO2	Apply data preprocessing, integration, transformation, and feature selection techniques using appropriate tools and libraries.
CLO3	Implement supervised and unsupervised data mining algorithms for classification, regression, clustering, and association analysis.
CLO4	Analyze mined patterns, model performance, and multidimensional data representations to derive meaningful insights.
CLO5	Evaluate and compare different data mining techniques and justify their suitability for real-world applications.

Course Outcome

	On completing the course, the student will be able to:
CO1	Design data warehouse schemas and implement fact and dimension tables with OLAP operations.
CO2	Perform ETL, data preprocessing, integration, discretization, and dimensionality reduction using tools such as Power BI, Python, WEKA, or R.
CO3	Develop predictive models using regression and classification algorithms and evaluate their performance using appropriate metrics.

CO4	Apply clustering and association rule mining techniques to extract useful patterns from datasets.
CO5	Conduct comparative performance analysis of multiple algorithms and interpret results for decision-making and knowledge discovery.

Sr No.	Data Mining and Warehousing Practical
1	Two case study on building Data warehouse/Data Mart Write Detailed Problem statement and design dimensional modelling (creation of star and snowflake schema)
2	Implementation of all dimension table and fact table based on experiment 1 case study using Appropriate schema
3	Implementation of OLAP operations: Slice, Dice, Rollup, Drilldown and Pivot based on experiment 1 case study
4	Perform different operations of extraction, transformation, and loading (ETL) processes on a sample dataset using PowerBI.
5	Integrate data from multiple sources by merging and transforming datasets using Python's pandas library and data manipulation techniques.
6	Apply feature selection techniques like variance thresholding and correlation analysis using Python's scikit-learn library to reduce dimensionality in a dataset.
7	Discretize continuous variables and create concept hierarchies for categorical variables in a market basket dataset using Python's pandas library.
8	Implement a linear regression method to make predictions based on the sample data set using Python.
9	Implement a logistic regression method to make predictions based on the sample data set using Python.
10	Build a decision tree classifier using Python's scikit-learn library to predict customer churn based on historical data.
11	Implementation of Clustering algorithm (K-means) in Python using scikit-learn to group customers based on their purchasing behavior.
12	Implementation of any one Hierarchical Clustering method.
13	Implement a Naive Bayes classifier in Python using scikit-learn to classify emails as spam or non-spam based on their content.
14	Implement the Apriori algorithm in Python to mine frequent itemsets from a retail transaction dataset and extract association rules.
15	To implement the FP-Growth algorithm in Python to discover frequent itemsets from a transactional dataset and analyze the results.
16	Perform Data Preprocessing Operations using WEKA / R Tool
17	Demonstrate Classification and Clustering Algorithms using WEKA / R Tool
18	Generate Association Rules using Apriori Algorithm in WEKA / R
19	Implementation of Page rank/HITS algorithm
20	Experimental Study and Comparative Performance Analysis of Multiple Classification Algorithms using Python

Course/ Paper Title	**No-SQL**
Course offered as	**VSC**
Course Code	**RUCSVSC605**
Semester	**VI**
No. of Credits	**2 (60 Hrs.)**
No. of lecture Hours/week	**4**

Sr No.	Course Learning Objectives:
CLO1	To understand the fundamentals and evolution of NoSQL databases.
CLO2	To compare RDBMS with different types of NoSQL databases.
CLO3	To provide hands-on experience with installation and usage of major open-source NoSQL databases.
CLO4	To perform data modeling using Document, Column, Key-Value and Graph databases.
CLO5	To develop applications using NoSQL databases.
CLO6	Develop a simple web/mobile application integrated with MongoDB using PHP or Python.

Course Outcome

	On completing the course, the student will be able to:
CO1	Explain the concepts, characteristics, and types of NoSQL databases.
CO2	Compare and contrast Relational Database Management Systems (RDBMS) with various NoSQL databases based on architecture, scalability, consistency, and performance.
CO3	Install the No-SQL Such as MongoDB, CouchDB.
CO4	Perform CRUD operations using open-source NoSQL databases such as MongoDB, CouchDB, Cassandra, Hive, HBase, Redis, Riak, Neo4j, and OrientDB.
CO5	Design and implement data models using Document-oriented, Column-oriented, Key-Value, and Graph databases for real-world applications.
CO6	Develop and integrate web/mobile applications using NoSQL databases and apply query processing techniques efficiently.

Sr No.	NoSQL Practical
1	Study and Introduction to NoSQL and types of NoSQL databases.
2	Introduction and Installation of MongoDB.
3	To write NOSQL queries to understand the concept of Open Source Database Management System such as MongoDB.
4	Perform data modelling with MongoDB.
5	Introduction and Installation of CoughDB.
6	To write NOSQL queries to understand the concept of Open Source Database Management System such as CoughDB.
7	Introduction and Installation of CASSANDRA.
8	To write NOSQL queries to understand the concept of Open Source Database Management System such as CASSANDRA.
9	Perform data modelling with CASSANDRA.
10	Introduction to Neo4j Graph Databases.
11	Basic Graph Queries and Implementations with Neo4j.
12	Introduction and Key-Value Operations on Redis.
13	To write NOSQL queries to understand the concept of Open Source Database Management System such as HIVE.
14	Introduction to OrientDB Graph Databases.
15	To write NOSQL queries to understand the concept of Open Source Database Management System such as OrientDB Graph.
16	To write a program for Mobile Database Query Processing using open source DB using MongoDB.
17	Introduction to HBase Installation.
18	To write NOSQL queries to understand the concept of Open Source Database Management System such as HBase.
19	Introduction and Key-Value Operations on Riak.
20	Design a web application using MongoDB (PHP/Python).

Semester V & VI Examination Pattern

Evaluation for Theory Courses (4 Credit Courses):

I. Continuous Internal Evaluation (CIA) - 40 Marks (40%)	II. Semester End Theory Examination (SEE)- 60 Marks (60%)
(i) Mid-Term Class Test – 20 Marks **(ii) Assignment/ Case study/ Presentations – 20 Marks**	• **Duration: 2 Hours** • **As per the paper pattern given below.**

Format of 4 Credit Theory question paper pattern:

Question	Based on	Options	Marks
Q.1	Module I	Any Three out of Six	15
Q.2	Module II	Any Three out of Six	15
Q.3	Module III	Any Three out of Six	15
Q.4	Module IV	Any Three out of Six	15
Total			60

● All questions shall be compulsory with internal choice within the questions.
● Each Question may be subdivided into sub questions as a, b, c, d, etc. & Each sub question carries 5 Marks.

Evaluation for Theory Courses (2 Credit Courses):

I. Continuous Internal Evaluation (CIA) - 20 Marks (40%)	II. Semester End Theory Examination (SEE)- 30 Marks (60%)
(i) Mid-Term Class Test – 10 Marks **(ii) Assignment/ Case study/ Presentations – 10 Marks**	• **Duration: 1 Hour** • **As per the paper pattern given below.**

Format of 2 Credit Theory question paper pattern:

Question	Based on	Options	Marks
Q.1	Module I	Any Two out of Four	10
Q.2	Module II	Any Two out of Four	10
Q.3	Module I & II	Any Two out of Four	10
Total			30

- All questions shall be compulsory with internal choice within the questions.
- Each Question may be subdivided into sub questions as a, b, c, d, etc. & Each sub question carries 5 Marks.

Major/Minor Practical Evaluation (50 Marks & 2 Hours Duration):

Sr. NO.	Practical Assessment	Marks
1	Practical Question 01(Module 1)	15
2	Practical Question 02(Module 2)	15
3	Viva	10
4	Journal	10

VSC & SEC Practical Evaluation (50 Marks & 2 Hours Duration):

Sr. NO.	Practical Assessment	Marks
1	Practical Question 01	20
2	Practical Question 02	20
3	Viva	5
4	Journal	5

- **Minimum 80% practical from each core subjects are required to be completed.**
- **Certified Journal is compulsory for appearing at the time of Practical Exam.**
- **Certified e-Journal (soft copy) is compulsory for appearing at the time of Practical Exam.**